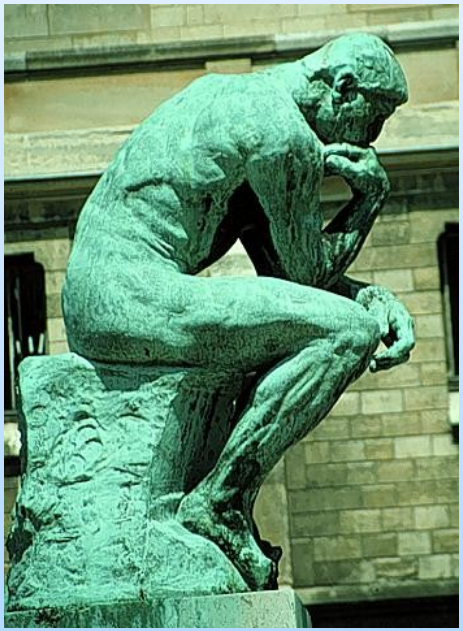
Introduction
to
Software Project Management

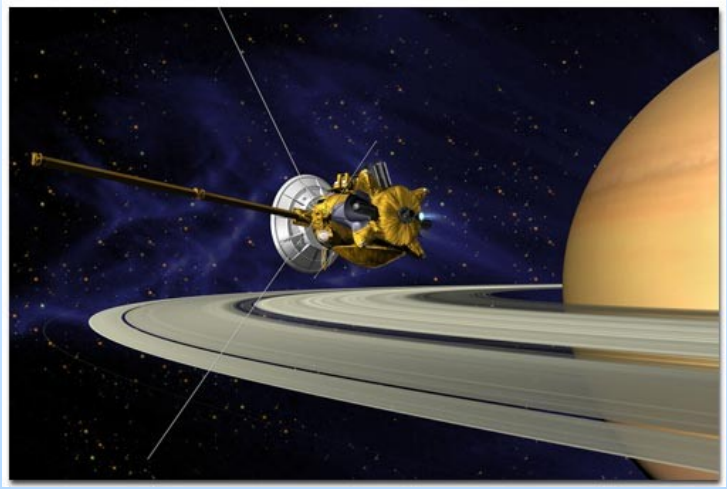
The Two End Points in a Project

Inspiration



A miracle occurs

Operation



INTRODUCTION/ FUNDAMENTALS

What is a Project?

A project is a temporary endeavor undertaken to create a unique product or service

- A Guide to the Project Management Body of Knowledge (PMBOK™), Project Management Institute, 2003

- **One time (non-routine)**
- **Limited funds/time (May be single phase or multiple phases)**
- **Specific resources utilized (may involve several specialism)**
- **Performed by people - Single or multi-person & multi location team(s)**
- **Planned, controlled (planning, monitoring and controlling is required)**
- **Specific Deliverables (specific objectives are to be met or a specified product is to be created)**

INTRODUCTION/ FUNDAMENTALS

Characteristics of projects

(From Bob Hughes and Cotterell) – addl points from exam point of view

- Non-routine tasks are involved
- Planning is required
- Specific objectives are to be met or a specific product is to be created
- The project has a predetermined time span
- Work is carried out for someone other than yourself
- Work involves several specialisms
- People are formed into a temporary work group to carry out the task
- Work is carried out in several phases
- The resources that are available for use on the project are constrained
- The project is large or complex

What is Management

- *Planning* – deciding what is to be done
- *Organizing* – making arrangements
- *Staffing* – selecting right people for the job etc.
- *Directing* - giving instructions
- *Monitoring* – checking on progress
- *Controlling* – taking action to remedy hold-ups
- *Innovating* – coming up with new solutions
- *Representing* – liaising with clients, users, developers, suppliers and other stakeholders.

The Management Spectrum

Effective project management focuses on four P's (in the order):

<u>The People</u>: Stakeholders, the team leaders, and the software team

- Deals with the cultivation of motivated, highly skilled people and teams
- Includes recruiting, selection, performance management, training, compensation, career development, organization and work design, and team culture development

<u>The Problem/ Product</u>: before a project can be planned

- Its objectives and scope should be established;
- Alternative solutions should be considered; and
- Technical and management constraints should be identified.

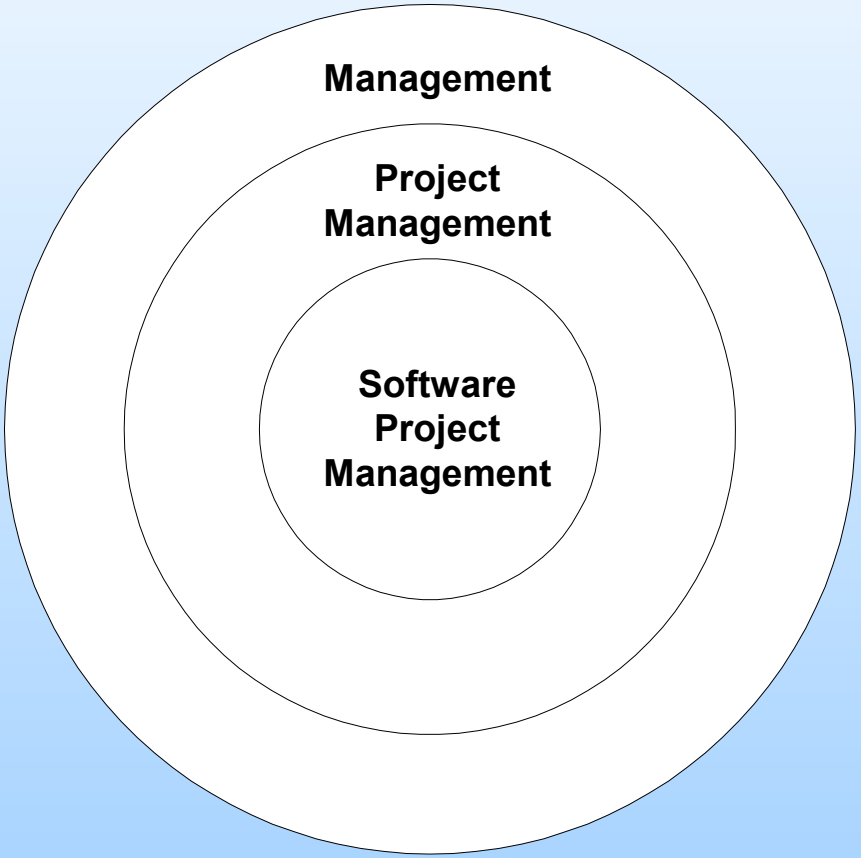
<u>The process</u>: a software process provides the framework from which a comprehensive plan for software development can be established.

<u>The project</u>: Planning and controlling a software project is done for one primary reason…it is the only known way to manage complexity

Project Management Skills

- Leadership
- Communications
- Problem Solving
- Negotiating
- Influencing the Organization
- Mentoring
- Process and technical expertise

Software Project Management



Software Project Management

- *Software project management* is aimed to ensure that the software is delivered on time, within budget and schedule constraints, and satisfies the requirements of the client

- Management of software projects is different from other types of management because:
  - Software is not tangible
  - Software processes are relatively new and still “under trial”
  - Larger software projects are usually “one-off” projects
  - Computer technology evolves very rapidly

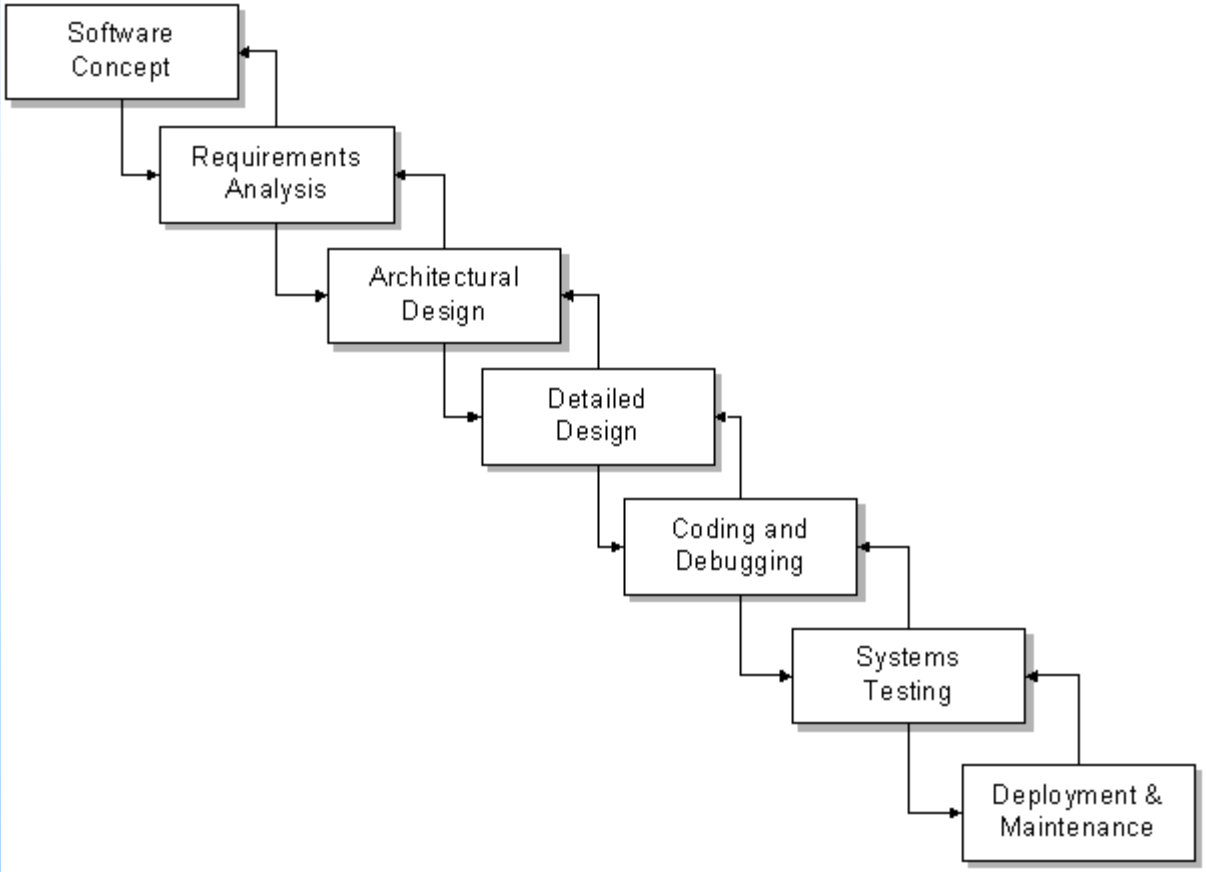
Software Projects versus other types of Projects

- *Invisibility* – With physical artifacts, measuring progress is easy as it can be seen/ felt. However with Software, progress is not immediately visible.
- *Complexity* – Software products are, generally, more complex than other engineering artifact of same value.
- *Flexibility* - It is easier to change/ modify software systems to meet changing organizational/ product requirement as compared to other engineering artifacts; it may not be possible to modify a physical artifact at all.

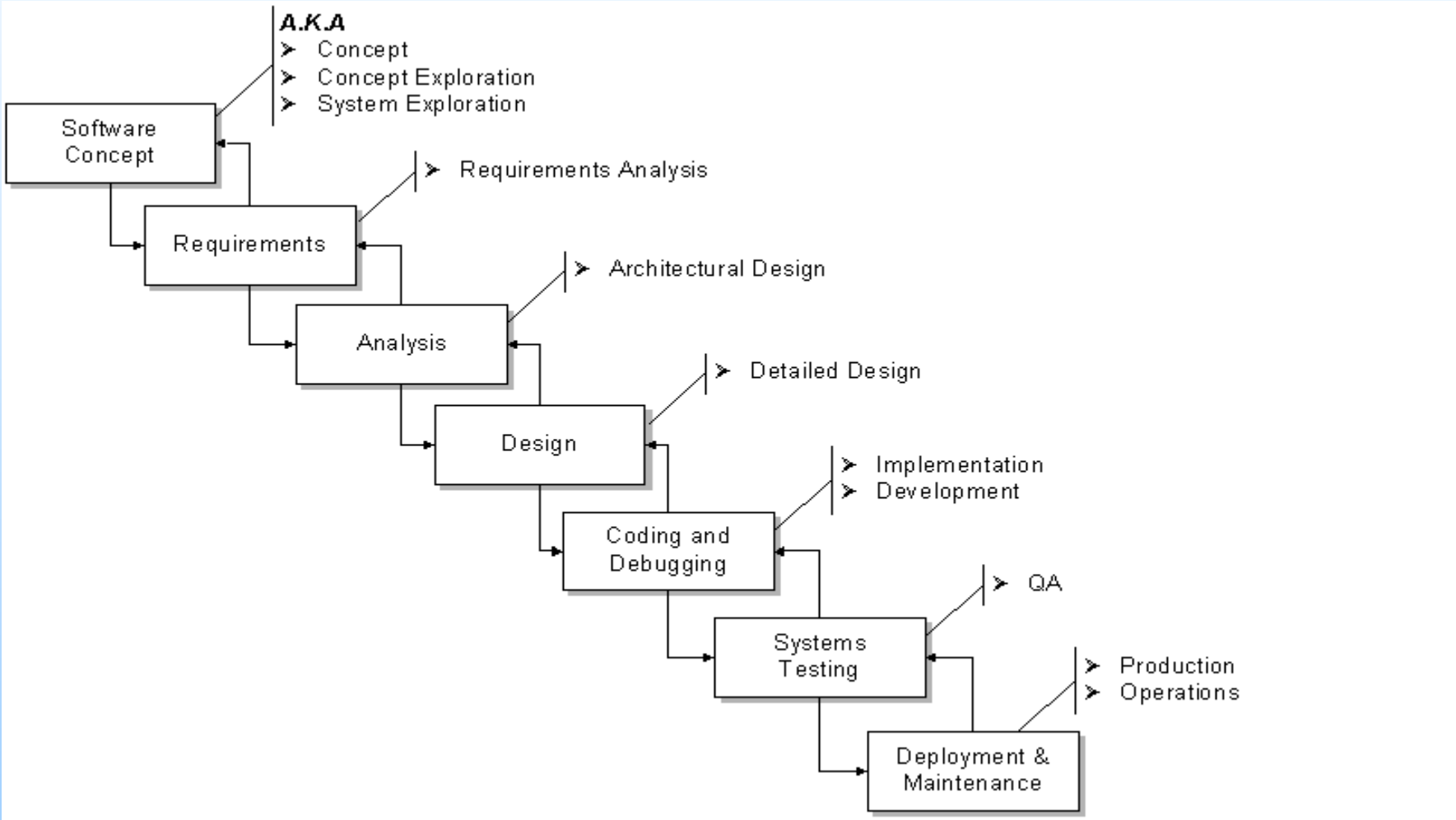
Project Phases

- All projects are divided into phases
- All phases together are known as the Project Life Cycle
- Each phase is marked by completion of Deliverables
- Identify the primary software project phases

Seven Core Project Phases



Project Phases A.K.A.



Interactions / Stakeholders

- As a PM, who do you interact with?
  - *Internal to Project team* – team members
  - *External to project team but within the same organization* - Project sponsor, Executives, Functional managers
  - *External to both the project team and the organization* – Customers, Contractors

Purpose of Project Management

- Ensure meeting the project objectives within the allocated schedule & budget
  - Communication
  - Meetings
  - Reviews
  - Authorization
  - Record Keeping
  - Monitoring (testing)
  - Interface Control
- Not for assigning blame (usually)

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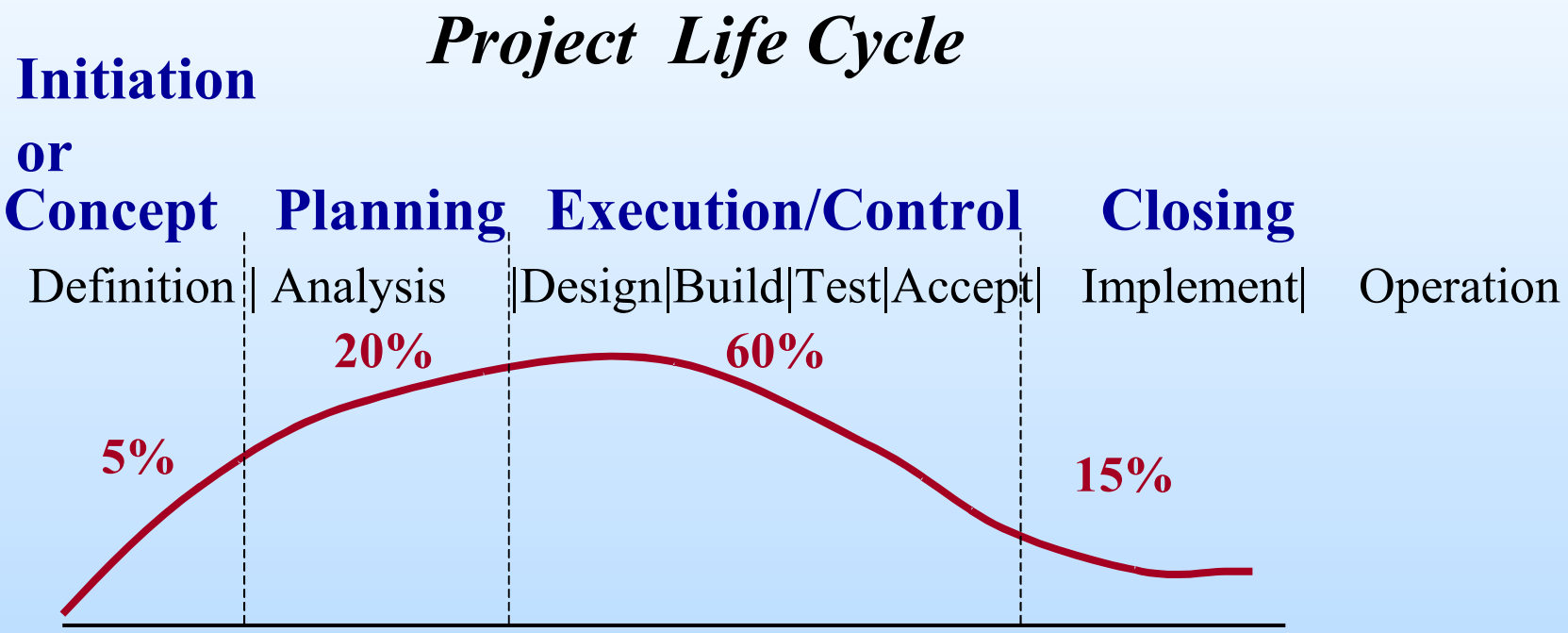


Percentages and graph refer to the amount of *effort (people)*
In IT projects = 90-95% of cost!

Project Life Cycle

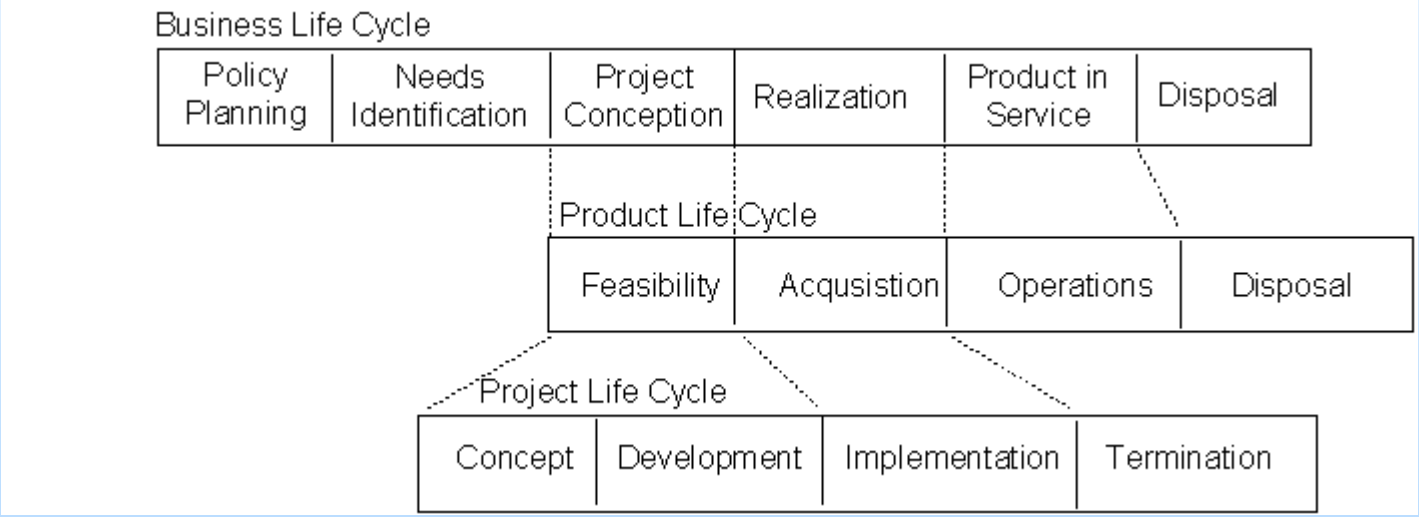
- **Initiation** – on the first stage, the necessity, feasibility, scope, time, budget and critical success factors of the project are defined along with the approach and methods to be used to deliver the required products and results.

- **Planning** – this stage includes a detailed identification of all the project elements and matters including project team, specified allocation of project resources and timeline, assignments of project tasks, evaluation of risks, definition of criteria for quality and successful completion of each deliverable, etc.

- **Execution** – this is the working phase where the project plan is implemented through practical actions that lead to successful project accomplishment. It is necessary to control performance and quality of all the required activities to know if they match the project's requirements.

- **Closure** – this stage identifies the project completion including testing, evaluation and formal acceptance of the final product by the customer, learning obtained from project experience, disposing resources, etc.

Lifecycle Relationships



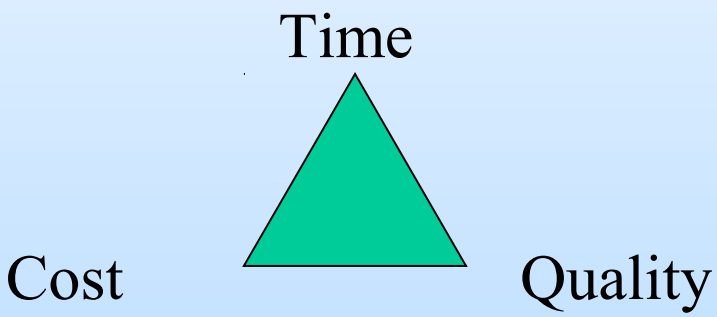
Project vs. Program Management

- What's a 'program'?
- Mostly differences of scale
- Often a number of related projects
- Longer than projects
- Definitions vary
- Ex: Program Manager for MS Word

INTRODUCTION/ FUNDAMENTALS

The Triple Constraint of Projects

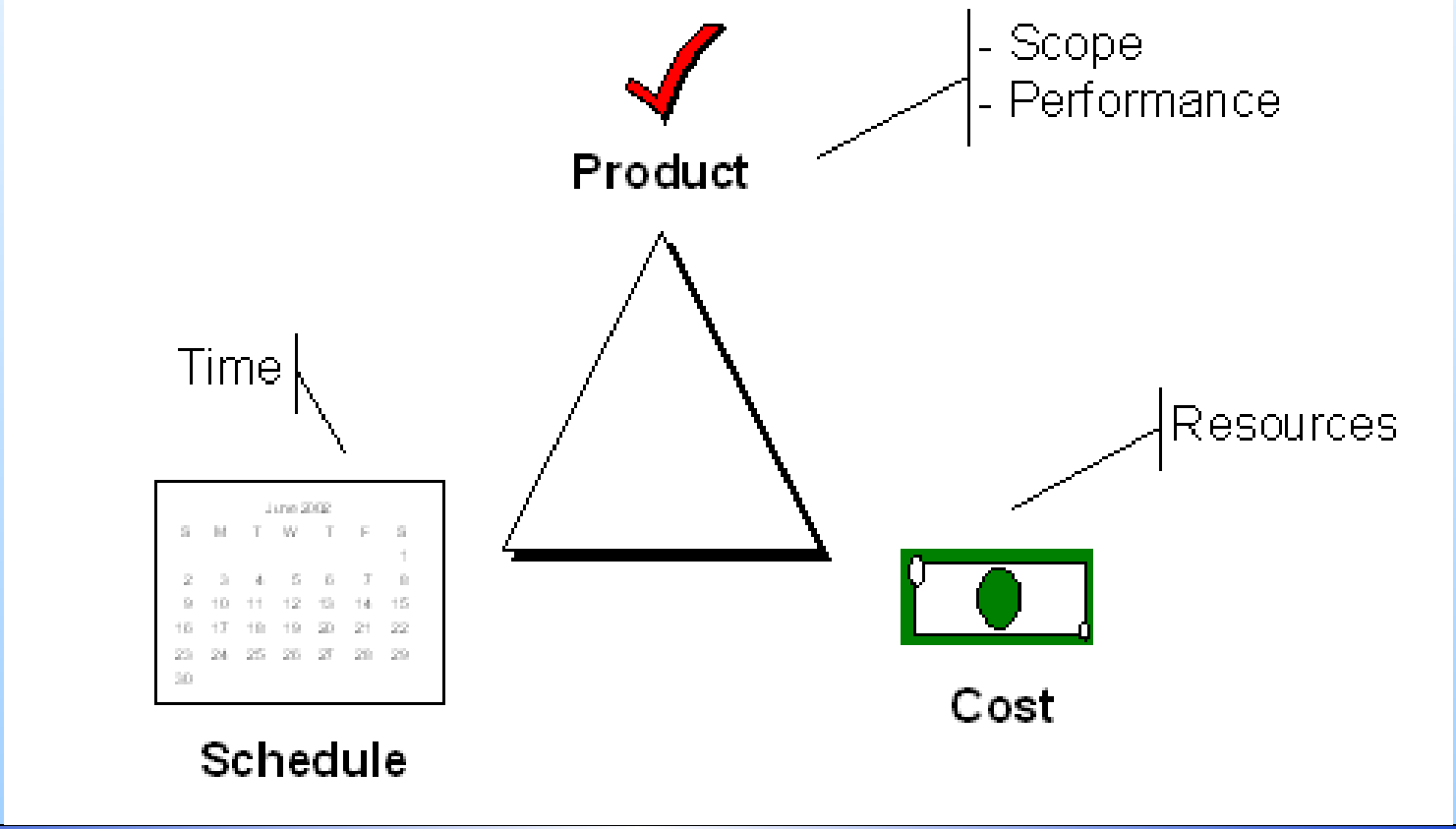
- **On Time, Budget, Quality = Required Scope**



- **Integration**
- **Trade – Off's**

Trade-off Triangle

- Know which of these are fixed & variable for every project



Problems with Software Projects

Manager's Point of View:

- Poor estimates and plans
- Lack of quality standards and measures
- Lack of guidance about making organizational decisions
- Lack of techniques to make progress visible
- Poor role definition – who does what?
- Incorrect success criteria

Members' Point of View:

- Inadequate specification of work
- Management ignorance of IT
- Lack of knowledge of application area
- Lack of standards

Contd/-

Problems with Software Projects

- Lack of standards
- Lack of up-to-date documentation
- Preceding activities not completed on time – including late delivery of equipment
- Lack of communication between users and technicians
- Lack of communication leading to duplication of work
- Lack of commitment – especially when a project is tied to one person who then moves
- Narrow scope of technical expertise
- Changing statutory requirements
- Changing software environment
- Deadline pressure
- Lack of quality control
- Remote management
- Lack of training

Assignment 1

Q1. What do you understand by Need identification, Vision and Scope for a product? Search on internet to find sample vision and scope document, copy the headings in your notebook and be prepared to present in the class –Group Activity – one per group.

Q2. What is the Vision Statement of ABES EC? What is the difference between Vision Statement and Quality Policy of an Organization? – Individual work.

Vision Statement

- **A vision statement** is sometimes called *a picture of your company in the future* but it's so much more than that. Your vision statement is your inspiration, the framework for all your strategic planning.
- A vision statement may apply to an entire company or to a single division of that company. Whether for all or part of an organization, the vision statement answers the question, "Where do we want to go?"

Vision Statement

General Motors – Vision Statement

"GM's vision is to be the world leader in transportation products and related services. We will earn our customers' enthusiasm through continuous improvement driven by the integrity, teamwork, and innovation of GM people."

Scope Definition

- A scope statement is one of the most critical pieces of a project, and writing one can be a difficult task for a project manager – no matter what type of project management methodology is being used. But, an effectively written scope statement can help the rest of the project flow along with minimal problems.

Scope Definition

The firsts step on writing a scope statement is filling in the *project name, project charter, and a listing of the project owner, sponsors, and stakeholders*.

Next, A *project justification* will need to be identified, as well as project *requirements, milestones, and deliverables.* (In scope Items)

Any non-goals - items that fall *outside of the scope* of the project - need to be identified here.

And finally, cost estimates need to be provided within the scope statement.

Note - This can be a cumbersome task, but it is a necessary one. As the project progresses, everyone involved knows where they can look should a question arise.

Questions?

Extra Slides

Four Project Dimensions

- People
- Process
- Product
- Technology

People-Related Mistakes Part 1

- Undermined motivation
- Weak personnel
  - Weak vs. Junior
- Uncontrolled problem employees
- Heroics
- Adding people to a late project

People-Related Mistakes Part 2

- Noisy, crowded offices
- Customer-Developer friction
- Unrealistic expectations
- Politics over substance
- Wishful thinking

People-Related Mistakes Part 3

- Lack of effective project sponsorship
- Lack of stakeholder buy-in
- Lack of user input

Process-Related Mistakes Part 1

- Optimistic schedules
- Insufficient risk management
- Contractor failure
- Insufficient planning
- Abandonment of plan under pressure

Process-Related Mistakes Part 2

- Wasted time during fuzzy front end
- Shortchanged upstream activities
- Inadequate design
- Shortchanged quality assurance

Process-Related Mistakes Part 3

- Insufficient management controls
- Frequent convergence
- Omitting necessary tasks from estimates
- Planning to catch-up later
- Code-like-hell programming

Product-Related Mistakes

- Requirements gold-plating
  - Gilding the lily
- Feature creep
- Developer gold-plating
  - Beware the pet project
- Push-me, pull-me negotiation
- Research-oriented development

Technology-Related Mistakes

- Silver-bullet syndrome
- Overestimated savings from new tools and methods
  - Fad warning
- Switching tools in mid-project
- Lack of automated source-code control

The Value of Project Management

- **Allows for excellent organization and tracking**
- **Better control and use of resources**
- **Reduces complexity of inter-related tasks**
- **Allows measurement of outcome versus plans**
- **Early identification of problems and quick correction**